Geldium Finance

Task 2: Predictive Modeling Plan

Predicting Credit Card Delinquency with GenAI Support

Purpose: Propose a transparent, testable approach for identifying customers who may become delinquent, so the Collections team can prioritize timely and appropriate outreach. This is a conceptual plan; no model has been trained and no predictive performance is claimed.

1 Generated Model Logic

Recommended initial model: logistic regression, with a decision tree as a comparison model. The target variable is `Delinquent_Account` (1 = delinquent, 0 = non-delinquent). Logistic regression estimates a probability between 0 and 1, which can be used to rank customers by estimated risk. A decision tree provides an interpretable comparison through decision rules.

Top five input feature groups proposed for review:

`Missed_Payments`
Recent missed-payment count may indicate repayment difficulty.

`Credit_Utilization`
Higher utilization may indicate greater pressure on available credit; confirm the dataset's scale first.

`Month_1`–`Month_6` payment history
Late or missed payment indicators may show repayment patterns, provided they precede the prediction date.

`Debt_to_Income_Ratio`
A higher debt burden relative to income may indicate less capacity to absorb expenses.

`Credit_Score`
A summary indicator of credit history that may help distinguish risk levels.

These are candidate predictors based on the data dictionary and credit-risk reasoning, not confirmed strongest predictors. Feature importance must be tested empirically. `Customer_ID` should be excluded because it is an identifier, not a meaningful risk signal. `Income` and `Loan_Balance` may also be evaluated as additional candidate features after missing-data handling.

Proposed workflow

1. Define the prediction point and target clearly: predict future delinquency using only information available at that time.
2. Review data quality from Task 1: investigate missing `Income` (39 records; 7.8%), `Loan_Balance` (29; 5.8%), and `Credit_Score` (2; 0.4%); standardize inconsistent `Employment_Status` labels; confirm whether `Credit_Utilization` is recorded as a fraction or percentage.
3. Split data into training and test sets using stratification so both sets retain a similar delinquency proportion. Keep the test set untouched until final evaluation.

4. Fit preprocessing only on the training data: impute missing numerical values using a justified method such as the median, encode categorical variables if used, and apply the same transformations to test data.

5. Train logistic regression as the baseline. Train a decision tree using the same split and comparable preprocessing.

6. Generate a delinquency probability for each customer. Use a threshold selected on validation data to assign a risk flag, and group customers into practical risk bands only after checking calibration and business capacity.

7. Evaluate on held-out data, review fairness and explanations, then recommend whether further refinement or a pilot is justified.

2 Model Choice and Justification

Logistic regression is a suitable starting point because `Delinquent_Account` is a binary outcome and the model produces a probability that can support risk prioritization. Its coefficients provide a relatively interpretable view of how input variables are associated with the predicted outcome, and the method is efficient to train, deploy, and monitor on structured customer data. This supports Geldium's need for explainable analysis and review by the Collections team.

Its limitations are that it may not capture complex nonlinear relationships without additional feature design, and its results depend on sound preprocessing, feature selection, and threshold choice. A decision tree should therefore be compared for interpretability and performance; the final model should be chosen from validation results rather than assumed to be best in advance. Neural networks are not the first choice for this initial exercise because the dataset is relatively small (500 records in the Task 1 EDA) and a more complex model may be harder to explain.

3 Evaluation Strategy

Task 1 EDA indicated approximately 16% positive delinquency cases. Because the target is imbalanced, accuracy alone can be misleading: a model that predicts every customer as non-delinquent could appear accurate while failing to identify delinquent customers. Use multiple metrics on a held-out test set.

Confusion matrix
Count true positives, true negatives, false positives, and false negatives to understand the types of errors.

Recall
Measure the share of actual delinquent customers identified. Low recall means the model misses customers who may need outreach.

Precision
Measure the share of customers flagged as delinquent who actually are delinquent. Low precision may create unnecessary outreach.

F1 score
Summarize the precision–recall trade-off when both false positives and false negatives matter.

ROC-AUC
Assess how well the model ranks delinquent customers above non-delinquent customers across thresholds.

Accuracy
Report overall correctness, but interpret it alongside the other metrics and the class distribution.

Calibration
Check whether predicted probabilities correspond to observed delinquency rates, especially if risk scores guide prioritization.

Fairness, bias, and explainability checks

- Compare recall, false-positive rates, precision, and flag rates across relevant customer groups where data and lawful use permit. Examine potential disparate impact rather than relying on one metric alone.
- Check for historical, selection, and proxy bias. Location and other variables should be reviewed carefully because they may act as proxies for sensitive characteristics.
- Use logistic-regression coefficients and, if appropriate, a validated explanation method such as SHAP to explain model behavior. Confirm explanations against the actual fitted model; do not rely solely on GenAI-generated rationales.
- Have human reviewers assess the proposed risk bands and outreach actions. A risk prediction should support responsible review, not automatically determine adverse customer decisions.

How results would guide improvements

If recall is too low, review the threshold and feature quality, and consider class-weighted training or carefully controlled resampling within the training data. If precision is too low, examine false positives and choose a threshold that fits the Collections team's capacity. If the model overfits or performs poorly, compare the decision tree, simplify features, or tune model settings using validation data. All final choices must be assessed on untouched test data. Do not claim an improvement unless measured results support it.

4 Responsible Use of GenAI

GenAI was used to structure the modeling workflow, compare candidate models, identify evaluation metrics, and draft fairness and explainability checks. Its suggestions are planning assistance, not evidence of model performance. Any future AI-generated code, interpretations, and recommendations must be reviewed and tested. Customer financial data should only be shared with tools approved for that data and use case.

5 Conclusion and Next Steps

Proceed with logistic regression as the baseline and a decision tree as the comparison. First resolve the identified data-quality questions, define the prediction timing to prevent data leakage, and select only features available at that time. Evaluate precision, recall, F1, ROC-AUC, the

confusion matrix, calibration, and fairness across relevant groups. Recommend a controlled pilot only if the held-out results, explainability review, and responsible-use checks are satisfactory.